

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a agile mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, Julia, MATLAB, C/C++, R, SQL
Tools: Linux, Git, GitHub Actions, Bash, PyTorch, OpenCV, open3d, Docker, Quarto, SSH, L^AT_EX

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- Developed [Python CLI tools](#)  for multispectral drone image processing using OpenCV; implemented multi-threading to cut processing times by 80% while managing version control.
- Contributed to the ongoing research effort to quantifying the land degradation on a scale using drone imagery analysis obtaining preliminary milestones of reliable calculation of vegetation index, and a method of quantifying weed percentage in the vegetation using spatial cluster analysis method.
- Replaced proprietary processing pipelines with open-source Docker application called [OpenDroneMap](#) , tuned the processing parameters and troubleshooted environment dependencies to ensure reliable output.
- Architected a unified, documented data storage structure for research backups, directly improving data retrieval efficiency and establishing a data management guidelines for future students.
- Engaged in field survey data collection and multispectral measurements; gained first-hand insight into ecological data collection to better design the drone imagery analysis pipeline with domain-specific knowledge.

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced machine learning demonstrations from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks with innovative interactive, reactivite programming environment using the [Pluto.jl](#)  package.

[Blog post: How to use TogetherAI in Julia](#)

- A quick Quarto blog post about using TogetherAI API in Julia programming language. Showcased a usecase of finding the most similar movie to a given movie using cosine similarity for the embeddings. Deployed onto the blog website using GitHub Actions.

AWARDS

Stipendium Hungaricum Scholarship Sep, 2025